

Section-5 Task Report

Overview:

This task introduces the fundamentals of combinational logic design and automated hardware verification using SystemVerilog.

- **The Design:** A hardware module was developed to simultaneously implement basic AND, OR, and XOR logic gates.
- **Automated Verification:** Instead of relying on manual waveform inspection, an automated "self-checking" testbench was created. It utilizes a for loop to exhaustively sweep through all four possible 2-bit input combinations (00, 01, 10, and 11).
- **Dynamic Checking:** For every input state, the testbench dynamically compares the physical hardware's output against a mathematically perfect software reference model.
- **Results:** The simulation completed successfully, generating a clean EPWave waveform and terminal logs that verified 100% logical accuracy by automatically printing a [PASS] message for every single test case.

EDA Playground Link: <https://www.edaplayground.com/x/pXfE>

design.sv:

```
`timescale 1ns / 1ps

module gates(a,b,y_and, y_or, y_xor);

    input logic a,b;
    output logic y_and, y_or, y_xor;

    assign y_and = a & b;
    assign y_or = a | b;
    assign y_xor = a ^ b;

endmodule
```

testbench.sv:

```
`timescale 1ns / 1ps

module gates_tb();

    logic a,b,y_and, y_or, y_xor;

    int i;

    logic [1:0] ab;
```

```

gates uut (.a(a), .b(b), .y_and(y_and), .y_or(y_or), .y_xor(y_xor));

task automatic check(input logic actual, expected, string name);
    if(actual !== expected)
        $display("[FAIL] %s: Expected = %0b Actual = %0b at Time = %0t",
name, expected, actual, $time);
    else
        $display("[PASS] %s: Value = %0b at Time = %0t", name, actual,
$time);
endtask

function automatic logic expected_and(input logic a,b);
    return a & b;
endfunction

function automatic logic expected_or(input logic a,b);
    return a | b;
endfunction

function automatic logic expected_xor(input logic a,b);
    return a ^ b;
endfunction

initial begin
    $display("--- Starting Self-Checking Exhaustive Test ---");

    $dumpfile("dump.vcd");
    $dumpvars(0, gates_tb);

    for(i=0; i<4; i++) begin
        ab = i[1:0];
        {a, b} = ab;
        #5;
        check(y_and, expected_and(a, b), $sformatf("AND (ab=%02b)", ab));
        check(y_or, expected_or(a,b), $sformatf("OR (ab=%02b)", ab));
        check(y_xor, expected_xor(a,b), $sformatf("XOR (ab=%02b)", ab));
        # 5;
    end
end

```

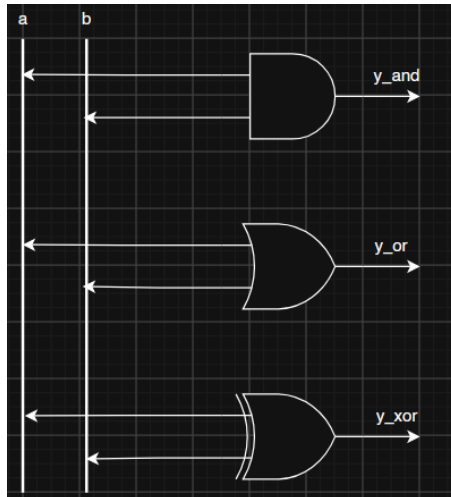
```

end

$display("--- Test Complete ---");
$finish;
end
endmodule

```

Schematic diagram:



Terminal Output:

```

--- Starting Self-Checking Exhaustive Test ---
[PASS] AND (ab=00): Value = 0 at Time = 5000
[PASS] OR (ab=00): Value = 0 at Time = 5000
[PASS] XOR (ab=00): Value = 0 at Time = 5000
[PASS] AND (ab=01): Value = 0 at Time = 15000
[PASS] OR (ab=01): Value = 1 at Time = 15000
[PASS] XOR (ab=01): Value = 1 at Time = 15000
[PASS] AND (ab=10): Value = 0 at Time = 25000
[PASS] OR (ab=10): Value = 1 at Time = 25000
[PASS] XOR (ab=10): Value = 1 at Time = 25000
[PASS] AND (ab=11): Value = 1 at Time = 35000
[PASS] OR (ab=11): Value = 1 at Time = 35000
[PASS] XOR (ab=11): Value = 0 at Time = 35000
--- Test Complete ---
$finish called from file "testbench.sv", line 48.
$finish at simulation time          40000

```

Waveform:

